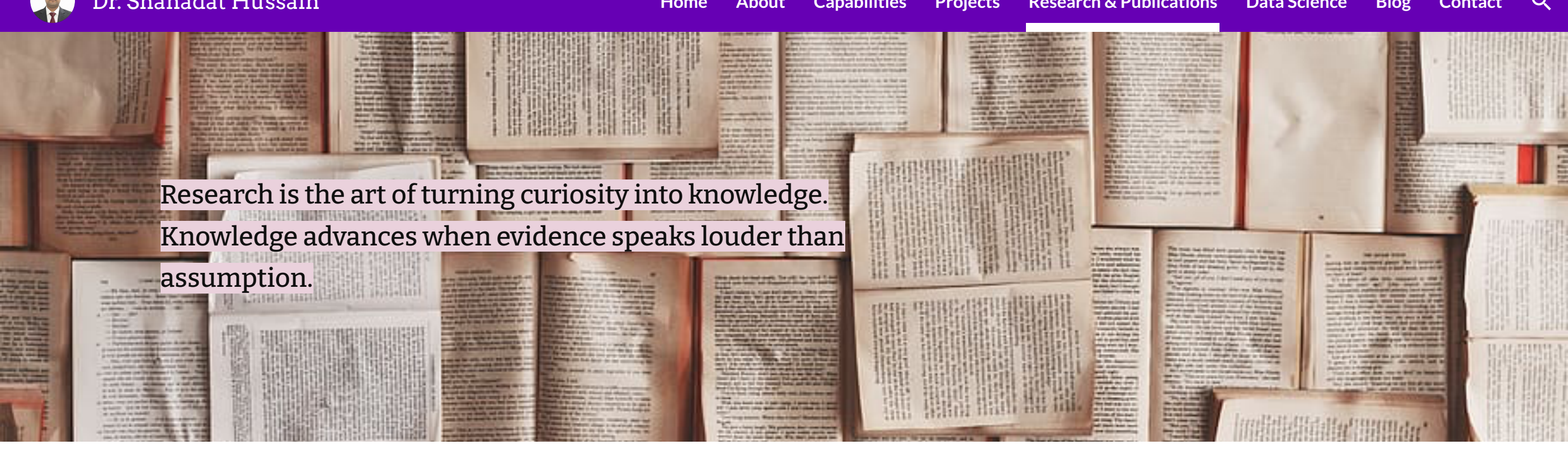

Research is the art of turning curiosity into knowledge. Knowledge advances when evidence speaks louder than assumption.

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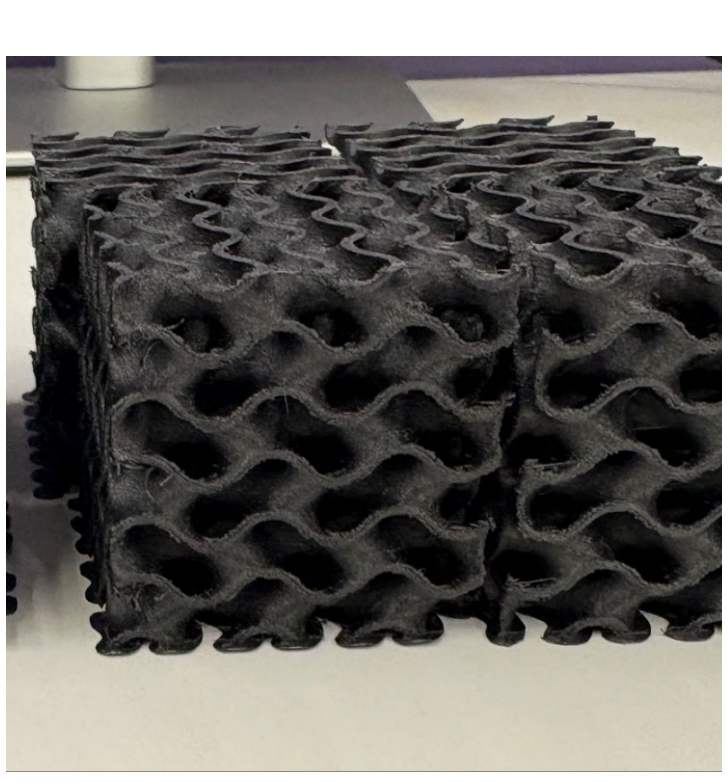
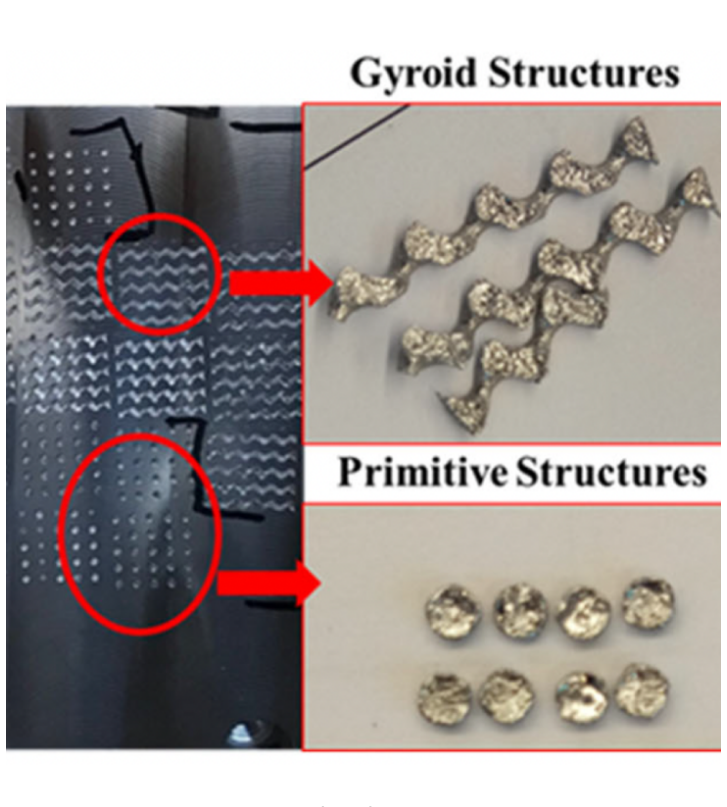
RESEARCH & ENGINEERING DOMAINS



FFF/SLA ▪ CF-PPA/PLA ▪ IPCs ▪ TPMS

New York University | 2024 - Present

- Do chopped fibers matter? Process-Structure-Property Relationships and Thermal Annealing Effects in Fiber-Reinforced TPMS Lattices
- Architected Fiber-reinforced polymer-matrix interpenetrating phase composites with tunable structural and damping attributes


Gyroid Structures
Primitive Structures

LPBF ▪ NiTi ▪ SMAs ▪ TPMS

Khalifa University | 2020 - 2024

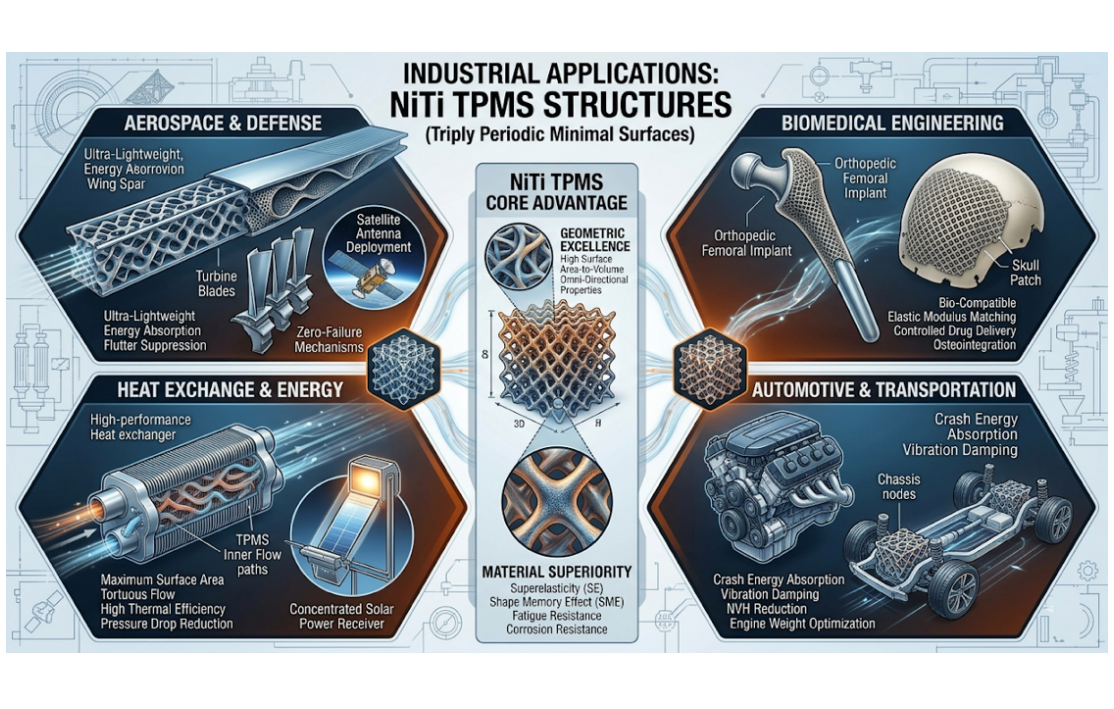
- Phase Transformation Behavior of NiTi Triply Periodic Minimal Surface Lattices Fabricated by Laser Powder Bed Fusion
- Microstructural and surface analysis of NiTi TPMS lattice sections fabricated by laser powder bed fusion
- Imperfections Formation in Thin Layers of NiTi Triply Periodic Minimal Surface Lattices Fabricated Using Laser Powder Bed Fusion

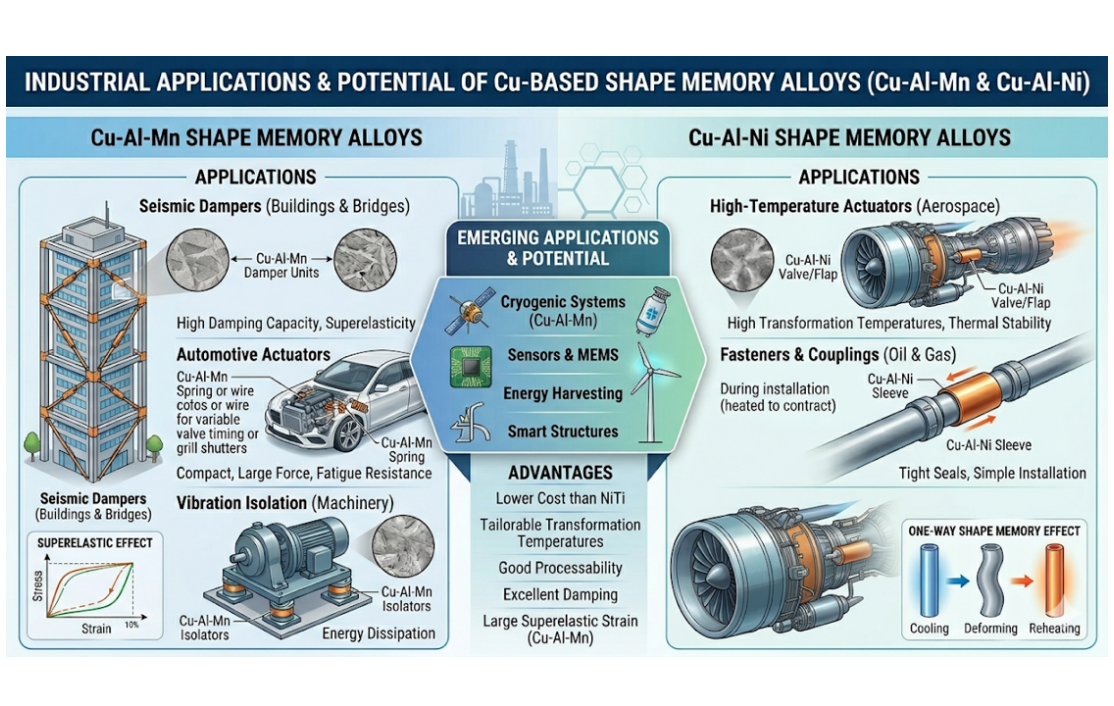
Induction Melting ▪ Cu-Al-Ni/Cu-Al-Mn ▪ SMAs

CSIR-AMPRI | 2013 - 2020

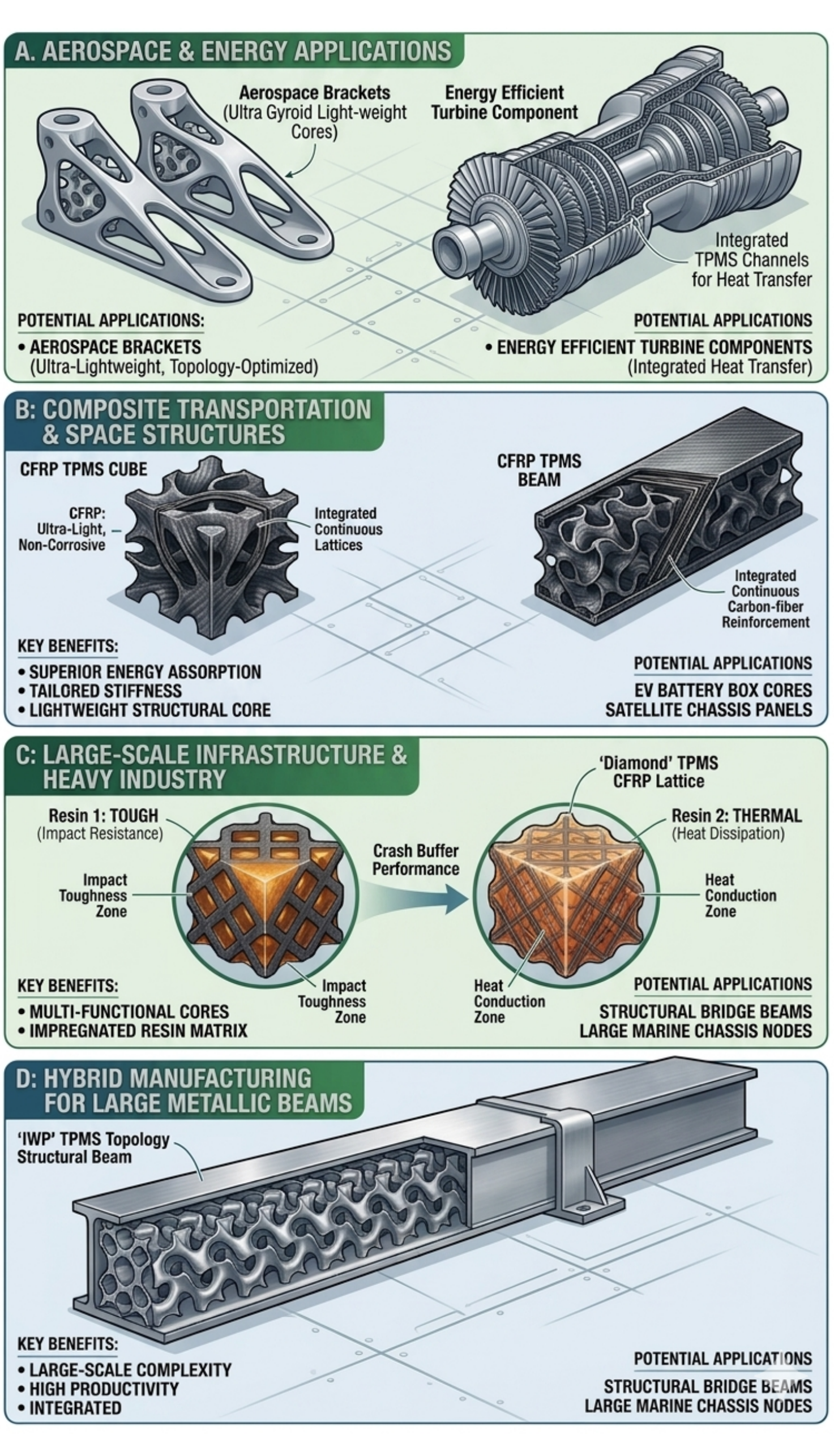
- Nano-CeO2 Doped Cu-Al-Ni SMAs with Enhanced Mechanical as well as Shape Recovery Characteristics
- Influence of niobium and silver on mechanical properties and shape memory behavior of Cu-12Al-4Mn alloys
- Designed polycrystalline ultra-high ductile boron doped Cu-Al-Ni based shape memory alloy

INDUSTRIAL APPLICATIONS


INDUSTRIAL APPLICATIONS:
NiTi TPMS STRUCTURES
(Triply Periodic Minimal Surfaces)
AEROSPACE & DEFENSE
BIOMEDICAL ENGINEERING
NiTi TPMS CORE ADVANTAGE
GEOMETRIC EXCELLENCE
HEAT EXCHANGE & ENERGY
AUTOMOTIVE & TRANSPORTATION
MATERIAL SUPERIORITY
Ultra-Lightweight Energy Absorption Wing Spar
Satellite Antenna Deployment
Turbine Blades
Zero-Failure Mechanisms
Orthopedic Femoral Implant
Skull Patch
Bio-Compatible Elastic Modulus Matching Controlled Drug Delivery Osteointegration
High-performance Heat exchanger
TPMS Inner Flow paths
Concentrated Solar Power Receiver
Crash Energy Absorption
Chassis nodes
Vibration Damping
NVH Reduction
Engine Weight Optimization


INDUSTRIAL APPLICATIONS & POTENTIAL OF Cu-BASED SHAPE MEMORY ALLOYS (Cu-Al-Mn & Cu-Al-Ni)
Cu-Al-Mn SHAPE MEMORY ALLOYS
Cu-Al-Ni SHAPE MEMORY ALLOYS
APPLICATIONS
Seismic Dampers (Buildings & Bridges)
High Damping Capacity, Superelasticity
Automotive Actuators
Compact, Large Force, Fatigue Resistance
Vibration Isolation (Machinery)
Energy Dissipation
EMERGING APPLICATIONS & POTENTIAL
Cryogenic Systems (Cu-Al-Mn)
Sensors & MEMS
Energy Harvesting
Smart Structures
ADVANTAGES
Lower Cost than NiTi
Tailorable Transformation Temperatures
Good Processability
Excellent Damping
Large Superelastic Strain (Cu-Al-Mn)
High-Temperature Actuators (Aerospace)
High Transformation Temperatures, Thermal Stability
Fasteners & Couplings (Oil & Gas)
Tight Seals, Simple Installation
ONE-WAY SHAPE MEMORY EFFECT
Cooling Deforming Reheating

- ➔ Aerospace & Defense
- ➔ Heat Exchange & Energy
- ➔ Biomedical Engineering
- ➔ Automotive Engineering
- ➔ Actuators
- ➔ Robotics
- ➔ Fasteners & Couplings
- ➔ Civil Engineering Structures


A. AEROSPACE & ENERGY APPLICATIONS
Aerospace Brackets (Ultra Gyroid Light-weight Cores)
Energy Efficient Turbine Component
Integrated TPMS Channels for Heat Transfer
POTENTIAL APPLICATIONS:
• AEROSPACE BRACKETS (Ultra-Lightweight, Topology-Optimized)
POTENTIAL APPLICATIONS
• ENERGY EFFICIENT TURBINE COMPONENTS (Integrated Heat Transfer)
B: COMPOSITE TRANSPORTATION & SPACE STRUCTURES
CFRP TPMS CUBE
CFRP: Ultra-Light, Non-Corrosive
Integrated Continuous Lattices
CFRP TPMS BEAM
Integrated Continuous Carbon-fiber Reinforcement
KEY BENEFITS:
• SUPERIOR ENERGY ASSORPTION
• TAILORED STIFFNESS
• LIGHTWEIGHT STRUCTURAL CORE
POTENTIAL APPLICATIONS
EV BATTERY BOX CORES
SATELLITE CHASSIS PANELS
C: LARGE-SCALE INFRASTRUCTURE & HEAVY INDUSTRY
Resin 1: TOUGH (Impact Resistance)
Impact Toughness Zone
Crash Buffer Performance
'Diamond' TPMS CFRP Lattice
Resin 2: THERMAL (Heat Dissipation)
Heat Conduction Zone
KEY BENEFITS:
• MULTI-FUNCTIONAL CORES
• IMPREGNATED RESIN MATRIX
POTENTIAL APPLICATIONS
STRUCTURAL BRIDGE BEAMS
LARGE MARINE CHASSIS NODES
D: HYBRID MANUFACTURING FOR LARGE METALLIC BEAMS
'IWP' TPMS Topology Structural Beam
KEY BENEFITS:
• LARGE-SCALE COMPLEXITY
• HIGH PRODUCTIVITY
• INTEGRATED
POTENTIAL APPLICATIONS
STRUCTURAL BRIDGE BEAMS
LARGE MARINE CHASSIS NODES

ALTMETRIC ATTENTION SCORE

Additive Manufacturing of Shape Memory Alloys: A Review with Emphasis on Powder Bed Systems

Ali N Alagha, **Shahadat Hussain**, Wael Zaki

Materials & Design, 204:109654, 2021

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🟥 **412** readers on Mendeley

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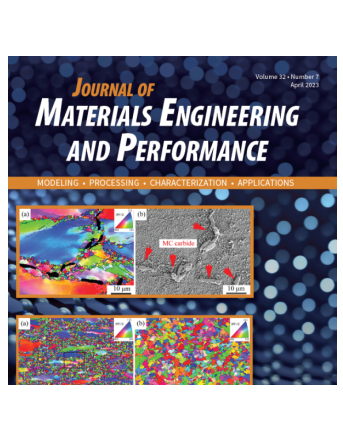
JOURNAL PUBLICATIONS



2024

Phase Transformation Behavior of NiTi Triply Periodic Minimal Surface Lattices Fabricated by Laser Powder Bed Fusion

Shahadat Hussain, Ali N Alagha, Wael Zaki

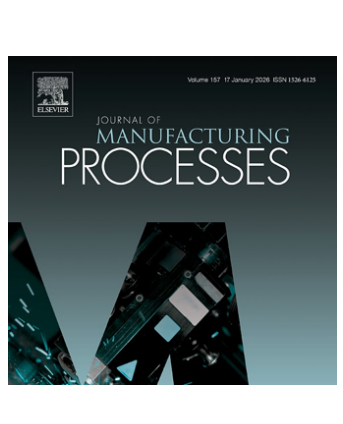
Journal of Materials Engineering and Performance, 2024

6

Citations

Abstract:

Laser powder bed fusion (LPBF) is an appealing additive manufacturing technique suitable for producing intricate products with complex shapes, where traditional subtractive manufacturing methods may be impractical. When dealing with NiTi shape memory alloys, the fabrication process encounters additional challenges due to the material's high ductility and work hardening characteristics. These properties lead to elevated tool wear and poor workability. However, LPBF overcomes these challenges by eliminating the need for tooling and enabling direct fabrication of complex shapes based on predefined computer-aided designs. Although there has been a growing interest in the LPBF of NiTi in recent years, the fabrication of NiTi structures with architected designs has received limited attention. This study aims to bridge this research gap by investigating the phase transformation behavior of two specific types of triply periodic minimal surface (TPMS) architected NiTi lattices, namely primitive and gyroid, manufactured using LPBF. The study utilized a methodology involving the design of TPMS structures, additive manufacturing, and the characterization of printed samples through electron microscopy, x-ray diffraction, and thermal analysis to measure phase transformation temperatures. The research not only examines the impact of process parameters on the behavior of the fabricated samples but also establishes the influence of cellular geometry on the functional response of TPMS NiTi structures. It was noteworthy to observe that gyroid samples displayed a higher thermal hysteresis and lower reverse transformation enthalpy in comparison with primitive samples.



2023

Microstructural and surface analysis of NiTi TPMS lattice sections fabricated by laser powder bed fusion

Shahadat Hussain, Ali N Alagha, Gregory N. Haidemenopoulos, Wael Zaki

Journal of Manufacturing Processes, 102:375-386, 2023

27

Citations

Abstract

Due to their high ductility and superior strength, the machining of NiTi shape memory alloys using conventional subtractive manufacturing technologies poses significant challenges. However, additive manufacturing (AM) offers a viable solution by circumventing the limitations of machinability through the elimination of tooling, thereby enabling the production of NiTi structures with previously unachievable intricacy. This study focuses on the fabrication of base layers of architected triply periodic minimal surface (TPMS) lattices using laser powder bed fusion (LPBF) and explores primitive and gyroid topologies. The investigation involves a comprehensive analysis and discussion of the influence of geometric properties and process parameters on the microstructural characteristics and the distribution of solid phases within the samples. Notably, the study reveals a substantial impact of process parameters and structural topology on the microstructural features. Additionally, notable observations are made concerning nickel evaporation, as well as the formation of oxide- and titanium-rich phases in relation to their distance from the base plate. Investigating intricate TPMS geometries in NiTi alloys in conjunction with varying laser process parameters represents a relatively new and unexplored area of research. These geometries may offer unique structural and functional properties that can potentially lead to innovative applications and advancements in various fields.

2022

PDF Available

Imperfections Formation in Thin Layers of NiTi Triply Periodic Minimal Surface Lattices Fabricated Using Laser Powder Bed Fusion

Shahadat Hussain, Ali N Alagha, Wael Zaki

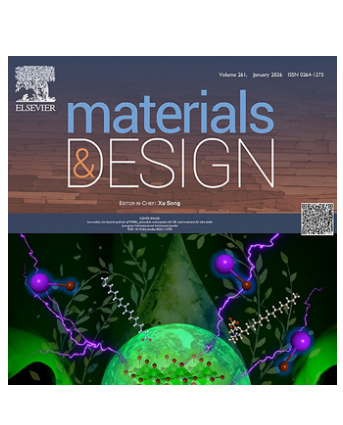
Materials, 15(22) 7950, 2022

11

Citations

Abstract

In this paper, thin layers of NiTi shape memory alloy (SMA) triply periodic minimal surface lattices (TPMS) are fabricated using laser powder bed fusion (LPBF), considering different laser scanning strategies and relative densities. The obtained architected samples are studied using experimental methods to characterize their microstructural features, including the formation of cracks and balling imperfections. It is observed that balling is not only affected by the parameters of the fabrication process but also by structural characteristics, including the effective densities of the fabricated samples. In particular, it is reported here that higher densities of the TPMS geometries considered are generally associated with increased dimensions of balling imperfections. Moreover, scanning strategies at 45° angle with respect to the principal axes of the samples resulted in increased balling.



2021

PDF Available

Additive Manufacturing of Shape Memory Alloys: A Review with Emphasis on Powder Bed Systems

Ali N Alagha, **Shahadat Hussain**, Wael Zaki

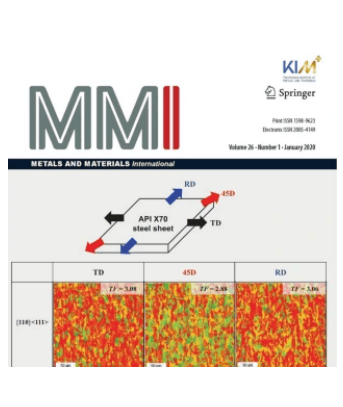
Materials & Design, 204:109654, 2021

179

Citations

Abstract

The ability of shape memory alloys (SMAs) to recover inelastic strains larger than any other metallic alloy has prompted their use in a wide range of applications. However, for the most common SMAs, including NiTi and Cu-based systems, fabrication using conventional means raises important challenges, including poor workability and potentially high tool wear. Additive manufacturing offers a direct answer to these challenges by eliminating the need for tooling and allowing the production of samples of complex geometries directly from computer-aided designs. The present work provides a comprehensive review of additive manufacturing applied to various SMA systems, with focus on the influence of process parameters and heat-treatment on the microstructure, printability, and the structural and functional behavior of additively fabricated samples.



2021

Nano-CeO2 Doped Cu-Al-Ni SMAs with Enhanced Mechanical as well as Shape Recovery Characteristics

Shahadat Hussain, Abhishek Pandey, Rupa Dasgupta

Metals and Materials International, 2021 27:1478–1482, 2021

4

Citations

Abstract

Effect of nano-CeO_2 doping was studied on self-accommodating Cu–12Al–4Ni based shape memory alloys. Cu–Al–Ni based alloys are known for their high thermal stability and good shape memory properties but mechanical processing of these alloys is very difficult due to their high brittle nature and susceptibility to inter-granular failure. The present work deals with the enhancement of mechanical as well as shape recovery characteristics of these alloys with the addition of nano-CeO_2. Significant improvements in ductility (1.8 times), strength and shape recovery were observed with the addition of nano-CeO_2 with respect to the base alloy.

2020

Influence of niobium and silver on mechanical properties and shape memory behavior of Cu-12Al-4Mn alloys

Abhishek Pandey, **Shahadat Hussain**, Prasanth Nair, Rupa Dasgupta

Journal of Alloys and Compounds, 2020

31

Citations

Abstract

The advantages of Cu-based shape memory alloys (SMAs) over the more popular Ni-Ti alloys are their low cost and easy processability coupled with good electrical and thermal conductivity. However, these alloys suffer from severe brittleness due to highly ordered structure causing difficulty in cold working; consequently, their applicability in the form of wires and sheets is impeded. In an attempt to improve the ductility of these alloys, varying amounts of Nb and Ag were added to Cu-12Al-4Mn, a known SMA, and the effect of these additives on the shape memory properties as well as the microstructure through grain refinement were analyzed. Detailed microstructural observations indicated that the addition of more than 2 wt% of Ag or Nb was detrimental to the martensitic structure that was formed on quenching. Furthermore, the addition of Nb led to the formation of a more desirable martensitic microstructure, namely the fine β_1' phase, whereas the inclusion of Ag caused the formation of the coarse γ'_1 phase. Based on the observed strength and shape memory effect, it was concluded that Nb significantly improved the mechanical as well as the shape memory properties of the base alloy.

2019

PDF Available

Effect of Microstructure on Roll-Ability and Shape Memory Effect in Cu-Based Shape Memory Alloys

Rupa Dasgupta, Abhishek Pandey, **Shahadat Hussain**, Ashish Kumar Jain, Ayub Ansari, V Sampath

Applied Innovative Research 1:29-37, 2019

?

Citations

Abstract

Ternary Cu-12Al-4Mn and Cu-12Al-4Ni have been chosen for the present study and different alloying elements have been added in an attempt to further improve upon the shape memory properties in terms of the desired microstructure in the quenched condition. It has been observed that all compositions do not exhibit the martensitic structure on quenching and different types of martensitic phases are exhibited depending on the alloying constituents. Attempts have been made to roll the samples into thin sheets through multiple passes. It has been found that although the ternary base alloys can easily be rolled the quaternary alloys with both Mn and Ni make the sample brittle rendering it unsuitable for rolling. Also adding of Zn and Mg has not exhibited the desired microstructure for exhibiting the shape memory properties and also could not be rolled though these samples were not brittle. The ease of roll-ability has been compared between the alloys and correlated with the microstructure which helps in understanding which type of martensitic phase is useful in rolling the samples.

2019

Designed polycrystalline ultra-high ductile boron doped Cu-Al-Ni based shape memory alloy

Shahadat Hussain, Abhishek Pandey, Rupa Dasgupta

Materials Letters 240:157-160, 2019

20

Citations

Abstract

Cu–Al–Ni shape memory alloys have good thermal stability and shape memory properties, but mechanical working of these alloys is very difficult due to inter-granular fracture and brittleness. This paper deals with improvement of shape memory properties and ductility of these alloys with the addition of boron. It was observed that with addition of boron in a designed experiment, ductility improved by more than four times than that of the base alloy. In addition, shape memory properties also improved significantly. These results obtained are discussed in this paper systematically.

2016

PDF Available

Effect of Nano CeO2 Addition on the Microstructure and Properties of a Cu-Al-Ni Shape Memory Alloy

Abhishek Pandey, Ashish Kumar Jain, **Shahadat Hussain**, V Sampath, Rupa Dasgupta

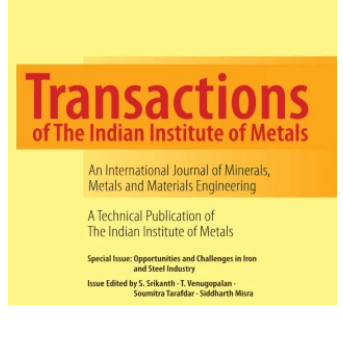
Metallurgical and Materials Transactions B 47:2205-2210, 2016

7

Citations

Abstract

This article deals with the effect of adding nano CeO_2 to act as a grain pinner/refiner to a known Cu-Al-Ni shape memory alloy. Elements were taken in a predefined ratio to prepare 300 g alloy per batch and melted in an induction furnace. Casting was followed by homogenization at 1173 K (900 °C) and rolling to make sheets of 0.5-mm thickness. Further, samples were characterized for microstructure using optical and electron microscope, hardness, and different phase studies by X-ray and transformation temperatures by differential scanning calorimetry. X-ray peak broadenings and changes were investigated to estimate the crystallite size, lattice strain, and phase changes due to different processing steps. A nearly uniform distribution of CeO_2 and better martensitic structure were observed with increasing CeO_2. The addition of CeO_2 also shows a visible effect on the transformation temperature and phase formation.



2015

PDF Available

Effect of Varying Al/Mn Ratio on Phase Transformation in Cu–Al–Mn Shape Memory Alloys

Ashish Kumar Jain, **Shahadat Hussain**, Pravir Kumar, Abhishek Pandey, Rupa Dasgupta

Transactions of Indian Institute of Metals 69:1289-1295, 2015

11

Citations

Abstract

In the present study, an attempt has been made to study the effect of the proportion of the main alloying constituents in a Cu–Al–Mn alloy, which is a known shape memory material. Four compositions of the alloy with varying ratios of Al:Mn, varying from 1 to 4 [added to copper], were synthesized using the liquid metallurgy route. After appropriate heat treatment to induce shape memory behaviour, they were studied for microstructure, X-ray diffraction, hardness and transformation temperature in an attempt to understand the effect of the varying ratios of the major alloying constituents on the properties mentioned. With an increase in the Al:Mn ratio, increase in grain size as well as cast hardness were observed. On the other hand, an increase in percentage decrease in hardness was observed with increase in Al:Mn ratio. Increase in Al:Mn ratio also favoured formation of martensitic structure with less amount of retained austenite.

2015

PDF Available

Effects of Different Quaternary Additions in the Properties of a Cu-Al-Mn Shape Memory Alloy

Shahadat Hussain, Pravir Kumar, Ashish Kumar Jain, Abhishek Pandey, Rupa Dasgupta

Materials Performance and Characterization 4:225-235, 2015

1

Citations

Abstract

In the present study, Cu-12.5 wt. %Al-5 wt. %Mn-80.5 wt. % shape memory alloy is chosen as the base alloy and 2 wt. % of quaternary additions of Fe, Ni, and Ti added to the base alloy. The effects of these additions on in terms of feasibility, microstructure, hardness, and transformation temperature were studied. The findings suggest the possibility to improve martensite formation, attain higher transitions temperatures, and longer retention over the base alloy through such additions.

2015

PDF Available

Changes in the properties of Cu-Al-Mn shape memory alloy due to quaternary addition of different elements

Pravir Kumar, Ashish Kumar Jain, **Shahadat Hussain**, Abhishek Pandey, Rupa Dasgupta

Revista Materia 20:284-292, 2015

27

Citations

Abstract

Advantages of Cu based shape memory alloy include amongst other features, high transformation temperature, low cost of production, ease in manufacturing processes and ability to vary the achieved properties through alloying additions. It has been often reported that these alloys are very sensitive to the alloying additions in terms of properties achieved and phase precipitation necessary for development of shape memory properties. This behaviour in Cu based shape memory alloys i.e. being very sensitive to its constituents can be used positively to design alloys with pre set properties if the alloying additions and their percentages are properly controlled. In an attempt to understand the effect of different alloying additions, 2% of different elements [Zn, Si, Mg & Cr] were added to a known Cu-based shape memory alloy [Cu-12.5 wt% of Al-5 wt % of Mn]. The objective was to ascertain changes or improvements achieved due to the additions in terms of microstructural changes, hardness, phase precipitation and transformation temperatures. Attempts have been made to analyze the changes in properties achieved in the base Cu-Al-Mn alloys due to the quaternary additions. Grain structure with α+β phases, which is a pre requisite for martensite formation on quenching is seen in all the alloys indicating that all the alloys have potential to exhibit the shape memory behaviour. The martensite formation with different morphologies is observed in the quenched samples however. XRD results have identified the precipitated phases to be the martensitic phases. The DSC results indicate clear transformation peaks in most of the samples with significantly high transformation temperatures. The findings confirm the variation in properties achieved due to different additions and improvements achieved in terms of higher transformation temperatures and martensite formation due to the alloying additions. An attempt has been made to understand the findings.

2015

PDF Available

Role of alloying additions on the properties of Cu–Al–Mn shape memory alloys

Rupa Dasgupta, Ashish Kumar Jain, Pravir Kumar, **Shahadat Hussain**, Abhishek Pandey

Journal of Alloys and Compounds 620:60-66, 2015

34

Citations

Abstract

The effect of alloying seven different elements [Zn, Si, Fe, Ni, Mg, Cr and Ti] on the microstructure, hardness, phase precipitation and transformation temperature in a Cu–12.5Al–5Mn alloy with a view to possible improvements as a result of these additions is the focus of the reported study. The base alloy has been chosen keeping in mind its ability to exhibit shape memory properties and improved ductility over other Cu-based SMAs. The objective was to ascertain changes or improvements attained due to the individual tertiary additions.

The samples were prepared through liquid metallurgy route using pure copper, aluminum, manganese and the respective quaternary alloying elements in right quantities to weigh 1000 g of the alloy in total and were melted together. Samples from the cast alloys were subject to homogenisation treatment at 200 °C for 2 h in a mufflefurnace and furnace cooled. Samples from the homogenised alloys were heated and held for 2 h at 920 °C followed by ice quenching to obtain the desired martensitic structure for shape memory behaviour. The alloys in the cast, homogenised and quenched conditions were metallographically polished to observe the martensitic phase formation mainly in quenched samples which is a pre requisite for exhibiting shape memory properties in these alloys. X-ray Diffraction studies were carried out on the cast and quenched samplesusing Cu Kα target; and the phases identified indicate martensitic phase precipitation; however in some cases the precipitation is incomplete. Differential Scanning Calorimetric [DSC] studies were carried out on quenched samples from room temperature to 600 °C maintaining a constant rate of 10 °C/min. Results indicate clear transformation peaks in all the samples which are significantly high than conventionally reported except with the addition of Mg in which case no distinct peaks have been recorded. The range of martensite retention is the maximum in ternary Cu–Al–Mn alloys; addition of quaternary elements decreases this range significantly. Presence of Ni delays austenite formation and completion [As and Af] significantly as compared to the ternary alloys; whereas with other additions the As and Af temperatures are brought forward. This means that whereas the alloys without quaternary additions would be better suited for its shape memory properties, ternary alloys would be better suited for higher transition temperatures.

The role of different alloying additions has been highlighted in the findings. Variations in properties have been attained due to different additions and improvements attained in terms of higher transformation temperatures and martensite formation due to the alloying additions.

2014

PDF Available

Effect of alloying constituents on the martensitic phase formation in some Cu-based SMAs

Rupa Dasgupta, Ashish Kumar Jain, Pravir Kumar, **Shahadat Hussain**, Abhishek Pandey

Journal of Materials Research and Technology 3:264–273, 2014

28

Citations

Abstract

The paper discusses the attempt made to improve upon the transition temperatures and amount of martensite formation in Cu–Al alloys with alloying additions of Mn, Ni and Zn in varying proportions and combinations. The alloys have been subjected to heat treatment cycles to improve upon the microstructure and precipitate the required martensite phase. The effect of these ternary and/or quaternary additions has been studied on the phases precipitated through micro-structural analysis and X-ray diffraction and Differential Scanning Calorimetric studies. The findings confirm the possibility of improvement on the shape memory properties like martensite formation, higher transitions temperatures and longer retention in selected alloys based on the properties exhibited through proper alloying additions and heat treatments.

CONFERENCE PUBLICATIONS

ASME IMECE 2022
Where Engineering Disciplines Intersect and Innovation Begins
Oct. 30 – Nov. 3, 2022
Greater Columbus Convention Center
Columbus, OH
event.asme.org/IMECE/

2022

Inhomogeneous Microstructure due to Non-Uniform Solidification Rate in NiTi Triply Periodic Minimal Surface (TPMS) Structures Fabricated via Laser Powder Bed Fusion

Shahadat Hussain, Ali N Alagha, Wael Zaki

SME International Mechanical Engineering Congress and Exposition (IMECE-2022), Ohio, USA, 2022

3

Citations

Abstract

In recent times, interest in the fabrication of porous NiTi structures have grown significantly. Porous structures have remarkable potential to be used in the areas of tissue engineering, impact absorption, and fluid permeability. However, fabrication of NiTi structures poses challenges such as poor machinability, high work hardening, and inherent springback effects, which render them difficult to tackle through conventional manufacturing routes. Additive manufacturing (AM) can alleviate the aforementioned issues associated with NiTi shape memory alloys (SMAs). In addition, this technology can be employed for producing metallic scaffolds and porous structures of complex architectural details. Recently, a class of minimal surface topologies, known as triply periodic minimal surface (TPMS) structures has emerged as an attractive configuration for building architected constructs. Very little work can be found in the literature addressing the fabrication of NiTi TPMS structures and investigating their behaviors. The complex geometries of these structures may influence the dynamics of the melt pool in beam-based AM processes as well as the solidification rate within different regions of a product, thereby affecting the microstructures of fabricated parts. An inhomogeneity in microstructures of fabricated parts was observed, which motivated a detailed examination of these structures. The novelty of the present work lies in studying the influence of geometries of NiTi TPMS lattices along with laser process parameters.

Advanced Materials Proceedings
Green and Sustainable World with Advanced Materials

2017

PDF Available

Study of effect of Fe, Cr and Ti on the martensite phase formation in Cu-12.5 wt%Al-5 wt%Mn SMA

Shahadat Hussain, Ashish Kumar Jain, Ansari MA, Abhishek Pandey, Rupa Dasgupta

Advanced Materials Proceedings 2(1):22-25, 2017

0

Citations

Abstract

Copper based shape memory alloys are studied throughout the world for their high transition temperatures and high thermal stability. Among Copper based shape memory alloys(SMAs), Cu-Al-Mn SMAs have shown good ductility and high transition temperature. Only those alloy systems that can show the formation of β phase are capable to demonstrate the shape memory properties. In this paper the effects of the alloying elements on the formation of martensite phase have been studied exclusively. Addition of 1 wt% of Fe, Cr and Ti to the Cu-12.5Al-5Mn shape memory alloy has been investigated in detail. Therefore, four alloys have been synthesized through liquid metallurgy route using pure metals of 99.9% purity in a melting furnace weighing 1kg each. Samples were heat treated at the temperature of 920˚C for 2 hours and then quenched in ice water. The optical micrographs show the formation of the martensite structures in all the samples except in the samples in which 1 wt% Fe was added. X-Ray diffractions also revealed the same facts as obtained in the optical microscopy. Vickers Hardness of all four samples were carried out. The result shows no sign of martensite formation in sample containing Fe; therefore, this alloy should not be used for further study in the direction of understanding shape memory behaviors of the copper based shape memory alloys. Moreover, it was also observed that the addition of Cr yielded good martensitic formation as compared to the alloy containing Ti.

ORAL PRESENTATIONS

BEST PRACTICES IN TEACHING AND LEARNING CONFERENCE

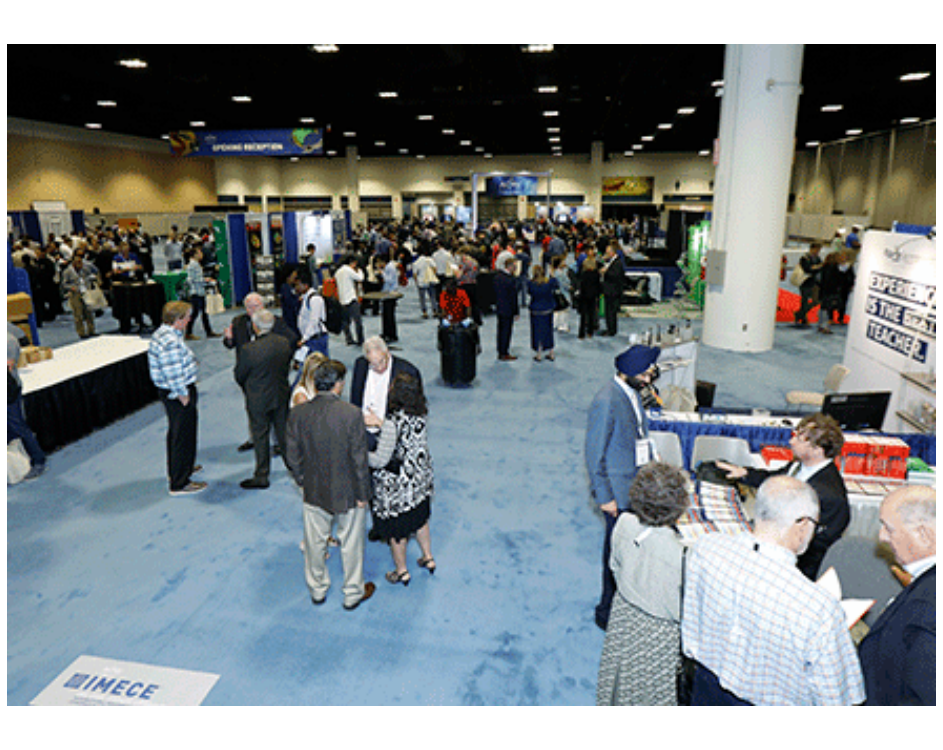
The Best Practices in Teaching and Learning Conference

BEST PRACTICES IN TEACHING AND LEARNING CONFERENCE
INSPIRING TEACHING AND ENRICHING LEARNING

2023

Venue: Khalifa University, May 2023

Organizers: Khalifa University, American University of Sharjah and Amity University Dubai



2022

Inhomogeneous Microstructure due to Non-Uniform Solidification Rate in NiTi Triply Periodic Minimal Surface (TPMS) Structures Fabricated via Laser Powder Bed Fusion

ASME-International Mechanical Engineering Congress and Exposition (IMECE-2022), Columbus, Ohio, USA November 2022

National Seminar on
Advances in
Smart & Functional
Materials
Date: 13-14th January 2017
Location: CSIR-AMPRI, Bhopal
CSIR-AMPRI

2017

Improvement in shape memory properties of CuAlNiZn shape memory alloys due to addition of grain refiners

National Seminar on Advances in Smart & Functional Materials held at CSIR-AMPRI, Bhopal on 13-14th January 2017. Organizers: CSIR-AMPRI Bhopal, IIM (Bhopal Chapter), MRSI (Bhopal Chapter) and TSI (Bhopal Chapter)

IAAI
International Assoc
Advanced Materials
Since 2010

2016

PDF Available

Study of effect of Fe, Cr and Ti on the martensite phase formation in Cu-12.5wt%Al-5wt%Mn SMA

International Conference on Materials Science and Technology held at University of Delhi, Delhi India on 1-4th March 2016. Organizers: International Association of Advanced Materials (IAAM), VBRI Press and University of Delhi

POSTER PRESENTATIONS

CSIR-AMPRI
RECENT INNOVATIONS IN ADVANCED MATERIALS (RIAM-2018):
PHYSICS OF ADVANCED MATERIALS
18-19th SEPTEMBER, 2018
HELD AT CSIR-AMPRI, BHOPAL
HELD AT
CSIR-AMPRI, BHOPAL

2018

Microstructural changes and its effect on mechanical as well as shape memory properties in a CuAlNi alloy on addition of dispersoid

Recent Innovations in Advanced Materials (RIAM-2018): Physics of Advanced Materials held at CSIR-AMPRI, Bhopal on 18-19th September, 2018.

Organizers: CSIR-AMPRI Bhopal, MRSI (Bhopal chapter), IIM (Bhopal chapter) and MPCOST Bhopal

SCIENCE FOR THE MASSES
INDIA INTERNATIONAL SCIENCE FESTIVAL 2016
National Physical Laboratory
New Delhi, India
7 - 11 December 2016

2016

Changes in properties due to alloying addition of Mn and Cr to Cu-Al-Ni-Zn shape memory alloys

India International Science Festival (IISF-2016) held at CSIR-NPL, New Delhi India on 7-11th December, 2016. Organizers: CSIR-NPL, CSIR-NISCAIR, Ministry of Science & Technology GOI, Ministry of Earth Sciences GOI, Vigyan Bharti and Vigyan Prasar

IIM
NMD ATM16

2016

Effect of Addition of Alloying Element Chromium to Cu-Al-Ni-Zn Shape Memory Alloys

70th IIM NMD-ATM Conference held at IIT Kanpur India on 11-13th November 2016.

Organizers: IIM, MSE-IIT Kanpur and DMSRDE Kanpur

NATIONAL SEMINAR
ON
RECENT INNOVATIONS IN
ADVANCED MATERIALS
29th December, 2015
CSIR-AMPRI, Bhopal, India

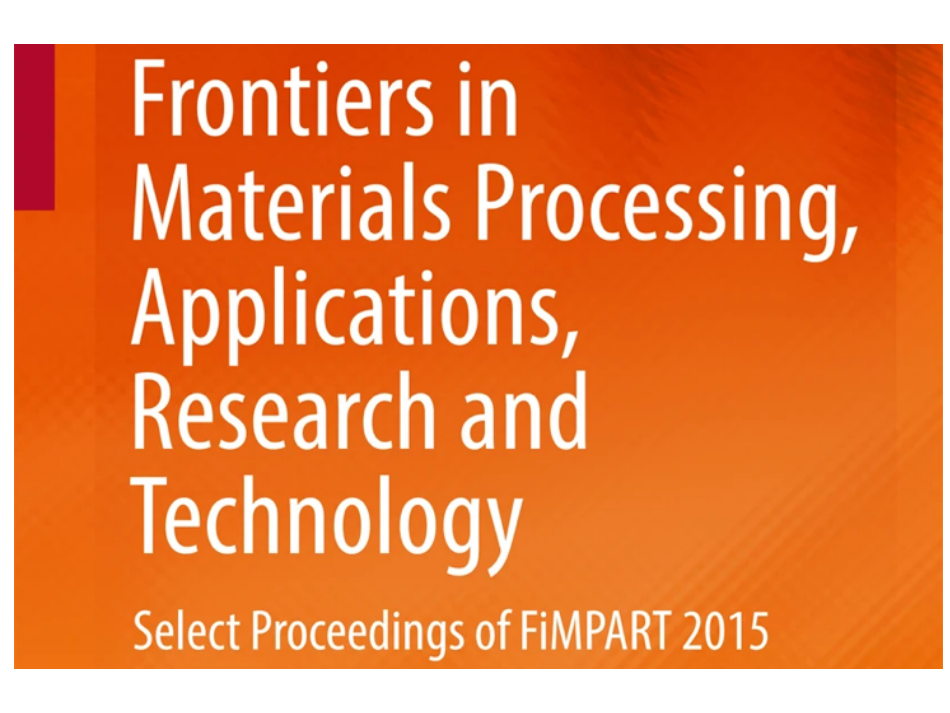
2015

Role of grain refiners for improved shape memory properties of Cu-12.5wt%Al-5wt%Mn Shape Memory Alloy

National Seminar on 'Recent Innovations in Advanced Materials' held at CSIR-AMPRI, Bhopal India on 29th December, 2015.

Organizers: CSIR-AMPRI Bhopal, MRSI (Bhopal Chapter) and MPCOST Bhopal

BOOK CHAPTER



2018

Effect of Alloying Additions on the Properties Affecting Shape Memory Properties of Cu–12.5Al–5Mn Alloy

3

Citations

Rupa Dasgupta., Jain A.K., **Shahadat Hussain**, Abhishek Pandey., V Sampath

In: Muruganant M., Chirazi A., Raj B. (eds) Frontiers in Materials Processing Applications, Research and Technology, Springer, Singapore, 2018

Abstract

The paper discusses the attempt made to understand the role of different alloying additions (Fe, Ni, Cr and Ti) to a Cu–12.5Al–5Mn alloy known to exhibit shape memory behaviour on its microstructure, phase precipitation and effect on transition temperatures after optimising the heat treatment cycles to precipitate the required martensite phase. The effect of the additions has been studied on the optimising conditions, phases precipitated through microstructural analysis, X-ray diffraction and transformation temperatures through Differential Scanning Calorimetric studies. In this study, the samples were prepared through liquid metallurgy route by melting pure metals. The cast alloys were subject to homogenisation treatment at 200 °C for 2 h in a muffle furnace and furnace cooled. In an attempt to precipitate the maximum amount of the desired microstructure of martensite in the quenched samples, the quenching cycle was optimised for each alloy by holding them for different duration of time from 30 to 120 min at 920 °C followed by ice quenching. X-ray diffraction studies carried out on the quenched samples indicate martensitic phase precipitation; however, in some cases, the precipitation is incomplete. Differential Scanning Calorimetric (DSC) studies carried out on quenched samples indicate clear transformation peaks in all the samples which are significantly higher than conventionally reported. The range of phase retention was used to determine the enthalpy and entropy changes. The findings confirm the possibility of changing the shape memory properties such as martensite formation, transitions temperatures, retention times, thermal properties as in entropy and enthalpy change with changing alloying constituents. Attempts have been made to study the shape memory properties of selected alloys that have exhibited promising improvement over the base alloy. The study can help pinpoint compositions with the desired properties in such alloy systems.

ARCHIVES & INDEXING

As with all peer-reviewed research, my work is automatically indexed across major scholarly databases; the corresponding selected records are listed here for completeness.

worldcat

Global Library Archive

- WorldCat (Online Computer Library Center-OCLC)
- DNB (German National Library)
- British Library
- HAL
- NYU Library
- KU Khazna
- PubMed Central (PMC)

scilit

Core Scientific Indexing

- Scopus (Elsevier)
- Web of Science (Clarivate)
- Google Scholar
- Scilit

openalex

AI-Powered Discovery Platforms

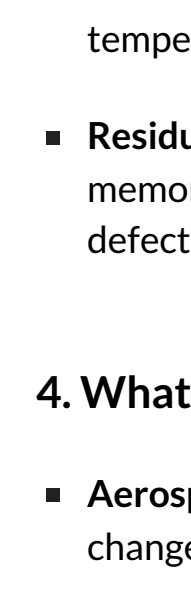
- Semantic Scholar
- OpenAlex
- Scispace
- Scholia
- Dimensions
- The Lens
- Sciprofiles

web of science

Verified Scholar Persistent Identifiers

- ISNI (International Standard Name Identifier)
- ORCID (Open Researcher and Contributor ID)
- Web of Science ResearcherID
- Scopus Author ID
- GND Number

FAQs



Frequently Asked Questions (FAQs)

1. What is the difference between 3D printing and 4D printing in your research?

While 3D printing focuses on creating a static three-dimensional object, 4D printing adds the dimension of time. In my work, I use "smart" materials like Nickel-Titanium (NiTi) Shape Memory Alloys. Once printed, these structures can transform their shape, stiffness, or volume when exposed to an external stimulus, such as heat or stress.

2. How do Triply Periodic Minimal Surface (TPMS) lattices contribute to 4D printing?

TPMS geometries, like the Gyroid and Primitive structures I study, are mathematically optimized for high surface area and energy absorption. When combined with 4D printing:

- **Programmable Response:** We can "program" how a lattice collapses or expands.
- **Functional Grading:** By varying the density of the lattice, we can create parts that react at different rates or temperatures across the same component.

3. What are the biggest challenges in 4D printing NiTi alloys via LPBF?

The primary challenge is maintaining the "memory" of the material. The intense heat of the laser in Laser Powder Bed Fusion (LPBF) can cause:

- **Nickel Depletion:** Minor changes in chemical composition that shift the transformation temperatures.
- **Residual Stress:** Internal stresses that can cause the part to warp or fail during its first "shape memory" cycle. My research specifically focuses on optimizing laser parameters to minimize these defects.

4. What industries benefit most from 4D-printed metamaterials?

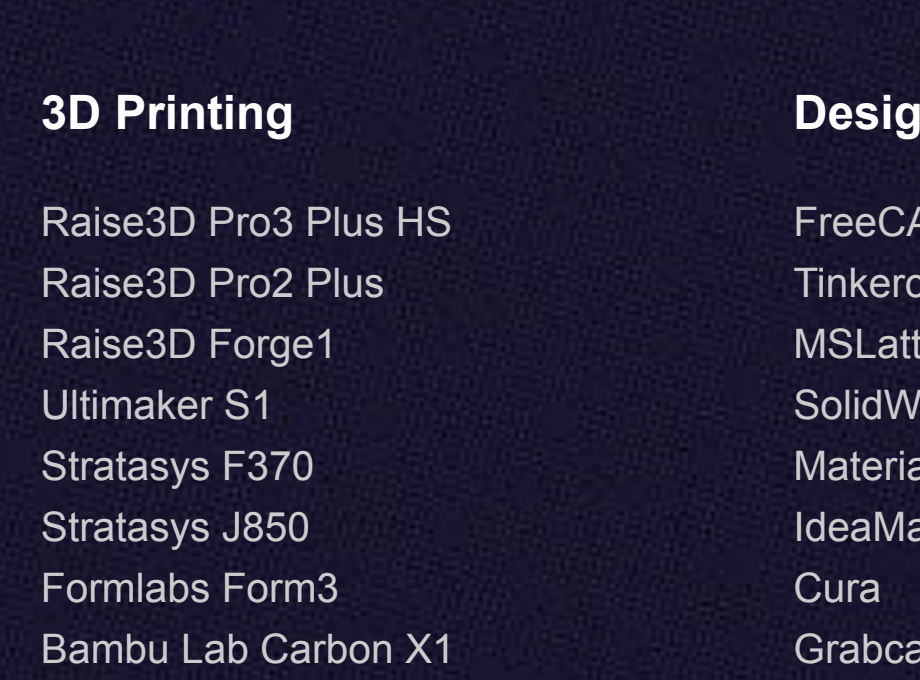
- **Aerospace:** Self-deploying solar panels or morphing wing structures that react to temperature changes without mechanical motors.
- **Biomedical:** Stents or bone scaffolds that are inserted in a compressed state and expand to their functional shape at body temperature.
- **Soft Robotics:** Actuators that mimic muscle movement through phase transformations.

5. Can 4D printing be applied to the polymer composites you work with?

Yes. Beyond metals, my work with Carbon Fiber-Reinforced Polymers (CFRP) can be adapted for 4D printing. By strategically aligning fibers during the print process, we can create "bimetallic-like" effects where the difference in thermal expansion between the polymer and the fiber causes the part to curve or fold when heated.

Explore my featured work related to research in metamaterials, TPMS lattices, 3D printing or additive manufacturing here ⇒

My featured work



Shahadat Hussain, PhD

Researcher in Metamaterials, 3D Printing & Shape Memory Materials

Exploring advanced materials for innovative applications

Key Profiles

- LinkedIn
- Google Scholar
- ORCID
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- The Lens
- GitHub
- Grokipedia
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3D Printing

- Raise3D Pro3 Plus HS
- Raise3D Pro2 Plus
- Raise3D Forge1
- Ultimaker S1
- Stratasys F370
- Stratasys J850
- Formlabs Form3
- Bambu Lab Carbon X1

Design & Simulation

- FreeCAD
- Tinkercad
- MSLattice
- SolidWorks
- Materialise Magics
- IdeaMaker
- Cura
- Grabcad Print
- PreForm
- Bambu Studio

Data Science

- MATLAB
- Python
- Pandas
- Numpy
- Matplotlib
- Plotly
- Seaborn
- Scipy
- Scikit-learn
- SQLite
- Jupyter Notebook
- Linux
- Shell Scripting
- LaTeX

Mechanical Testing

- Instron 5969 (Bluehill Universal)
- Instron 8872 (WaveMatrix3)
- MTS 100 kN (TestSuite TW)
- Controls Simplex 350 & Automax (Multitest)
- KB 250 BVRZ (HardWin XL)

AVAILABILITY
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Business Card

Micro-Credentials

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